GM Campus Power Quality — Vision, Ideas & Blueprint

Oct 12, 2026

Understanding the project: what is expected

By the end of the winter term, the team must deliver a system running on real campus meters that facility management finds useful, plus a report and a presentation.

The project in plain words

The Gummersbach campus is installing several electricity meters. Each one works alone today, so nobody sees the whole picture. Our job is to build one system that brings all of them together.

Think of it as a fitness tracker for the campus's electricity. A smartwatch does not just show your heart rate: it records it all day, spots irregularities, sends a weekly summary and warns you when something looks wrong. Our system should do the same for the campus power supply.

Who is involved

Who	Role in the project
Prof. Dr. Michael Freiburg, Prof. Dr. Felix Hackelöer	Supervisors: they grade the work and set the scope
TH Köln facility management	The real customer: owns the electrical installation, gives access, and uses the system after we leave
Krishnapriya, Hanna, Gaurav	The team that builds it

What the work boils down to

1. **Collect**: get the data out of every meter.
2. **Store**: keep it safely in one central place.
3. **Analyse**: turn the numbers into answers.

4. **Show**: present those answers to the people who need them.

The dashboard is only step 4. "Not just a dashboard" means steps 1 to 3 matter just as much. Step 3 is where we can stand out: not "here is the voltage" but "here is what happened, why, and what to do".

Why this project matters

Today every meter on the GM campus is an island. This project turns them into one trusted system that shows how much power the campus uses, how clean that power is, and what to do about it.

The problem

- Several measurement devices are being installed on the Gummersbach campus, but each one works on its own. Nobody sees the campus as a whole, and data that is never collected centrally is effectively lost.
- Building loads have changed. LED drivers, PCs and servers, variable-speed drives in ventilation and lifts, EV chargers and PV inverters are all power electronics.
- Power electronics draw non-sinusoidal current. The result is harmonics, flicker, unbalance and transients that older building installations were never designed for.

Who gains what

Stakeholder	What they gain	Concrete example
TH Köln facility management	Visibility of every feeder, and recorded evidence for fault-finding	A breaker trip is matched to a voltage dip recorded in the same second
University management	Measured data for its legal energy-saving duties	EnEfG § 6: public bodies using 1 GWh or more per year must cut final energy use by 2 % per year until 2045 and run an energy-management system
Electrical assets and safety	Early warning before damage	Triplen harmonics (3rd, 9th, 15th) add up in the neutral conductor and can overheat it

Stakeholder	What they gain	Concrete example
Energy budget	Knowing, and shaving, the yearly peak	Large German customers with load-profile metering usually pay a demand charge set by their highest 15-minute average power
Our team	The full IT/OT chain from sensor to decision	The Industrial IoT profile companies hire for

TH Köln as a whole almost certainly uses more than 1 GWh a year. Ask facility management how it meets the EnEfG duties today: our system could feed straight into that.

What we can gain

Beyond the grade, each of us ends the semester with a working system on real campus infrastructure to show for it.

What we gain	What it means in practice
The official learning outcomes	Power and power-quality measurement in a large building, collecting data from the field, and evaluating real measurement data
The full IT/OT chain	From a meter on the wall to a decision on screen: communication, data pipeline, database, analysis and dashboard
Experience with real, messy data	Gaps, wrong timestamps and odd values, which textbook datasets never have
Working with a real customer	Finding out what facility management needs, and working within safety rules, IT approvals and deadlines
Teamwork and project management	Planning a semester, splitting roles, documenting, and presenting to professors and facility management

How we will do this project (Timelines)

Build one thin slice that works end to end first, then make it wider and smarter. Never build everything for every meter at once.

The order of work

1. **Discover (October).** Meet the professors and facility management. Walk the campus with them and list every meter: where it is, which model, and how it communicates. Look at the data we were given. Request network access now, because IT security issues. What type and how is the dtaabase provided.
2. **Plan (late October to early November).** Agree on what the system must do, sketch a simple architecture.
3. **End to end (November).** Collect, store and show data for visualisation and some additional features. Ugly but working is fine: it proves the whole chain early, while there is still time to fix surprises.
4. **Grow (December).** Connect all meters and make the system reliable: catch missing data, survive network outages, and keep timestamps correct.
5. **Add the brain (January).** Power-quality evaluation, weekly reports and alerts. This is the "not just a dashboard" part where we stand out.
6. **Finish (late January to February).** Test, document, hand over to facility management, write the report and prepare the presentation.

Innovation: going beyond the dashboard

The dashboard shows what is happening. Our innovations explain why it happened, warn before problems, and save facility management work.

How innovations plug into the system

Every idea below follows the same pattern. A small program reads from the database on a schedule, works something out, and writes the result back: to the dashboard, a report or an alert. So once the end-to-end slice works, each innovation is just one more small program.

Ideas and how we would build them

Idea	What it gives the campus	How we would build it
Data health check	Trust: we know a number is wrong before anyone acts on it	Simple rules run every few minutes: is data missing, is a value frozen, do the sub-meters add up to the main meter? Problems appear as flags on the dashboard.
Weekly power-quality report	A ready-made summary for facility management instead of digging through charts	Every Monday, a script compares last week's values with the EN 50160 limits and fills a short report that is e-mailed out.
Event detective	Tells whether a voltage dip came from the public grid or from inside the campus, and where	Collect the events each meter records and line them up by time. Seen by all meters means the grid; seen by one means inside the campus. Needs accurate clocks on all meters.
Campus context and waste finder	Shows energy used when buildings are empty: nights, weekends, holidays	Add the lecture timetable, holidays and weather to the power data, then compare empty hours with busy hours.
Smart alerts	Warns when something unusual happens, such as ventilation running all Sunday	Start by comparing each hour with the same hour in previous weeks. Later, a machine-learning model learns what "normal" looks like. Alerts go out by e-mail or Teams.
Peak warning	Warns before the campus sets a new yearly power peak, which drives cost	Forecast tomorrow's load from history, weather and the timetable, and warn if the peak looks too high. Needs several weeks of history; the dataset we were given helps.

Each idea in detail

1. Data health check

What it means: the system checks its own data before anyone uses it, like a smoke detector for bad data.

Example: a meter's network cable is unplugged on Friday evening. Without a check, the dashboard shows a flat line, and someone might think the building used no power all weekend. With the check, the dashboard says "Meter 3: no data since Friday 18:00".

How it works: it checks four things:

- **Missing data:** gaps in the readings.
- **Frozen values:** the same number for hours. Real electricity always moves a little.
- **Impossible values:** for example, 500 V on a 230 V system.
- **Balance:** do the sub-meters add up to the main meter? If not, a meter is wrong or some load is not metered.

Facility management sees a simple green, yellow or red status for each meter.

2. Weekly power-quality report

What it means: an automatic health report on last week's electricity quality, arriving every Monday.

Example content: "Voltage stayed within limits all week. Distortion was highest on Wednesday at 11:00 in Building B. Two short voltage dips occurred on Tuesday."

How it works: EN 50160 sets limits, such as voltage within ±10 % for 95 % of the week. The program checks last week against those limits, writes a one-page summary with a few charts and a pass or fail verdict, and e-mails it.

Why it matters: facility management has no time to browse dashboards, so the report comes to them.

3. Event detective

What it means: when the voltage briefly drops, it works out where the problem came from.

Example: on Tuesday at 10:42 the PCs in a lab rebooted. Was it the public grid, or something inside the campus?

- If every meter saw the dip at the same moment, it came from outside.

- If only one sub-distribution saw it, the cause is inside that part of the building, such as a large machine starting.

How it works: the meters record events with timestamps. The program lines them up by time and compares them.

What it needs: meters that record events, and clocks synchronised across all meters.

4. Campus context and waste finder

What it means: power data alone says how much was used. Adding campus life says whether it was necessary.

Example: a building draws high power at 3 a.m. on a Sunday, when nobody is there. Lights, ventilation or equipment were left running.

How it works: the program adds the lecture timetable, holidays and weather to the power data. Then it compares empty hours with busy hours, using measures such as night and weekend base load or energy per lecture hour.

Machine learning: clustering groups days by their shape, such as "normal lecture day" or "weekend", and spots odd days, such as a Sunday that looks like a Monday.

5. Smart alerts

What it means: the system notices unusual behaviour and tells someone straight away, not at the end of the month.

Example: the ventilation runs all Sunday because a timer failed. Facility management gets an e-mail that morning instead of a surprise on the bill.

How it works:

- **Simple version:** compare each hour with the same hour in previous weeks.
- **Machine-learning version (Isolation Forest):** learns normal patterns across many signals at once and flags combinations it has never seen.

Key point: every alert says why it fired, and false alarms stay rare. Otherwise people start ignoring the alerts.

6. Peak warning

What it means: large electricity customers usually pay a charge based on their single highest 15-minute peak of the year. One bad moment can raise the bill for the whole year.

Example: on a cold January morning, lectures start at 8:00 and the heating, ventilation and all the labs switch on together. That sets a new peak.

How it works: a machine-learning model such as LightGBM forecasts tomorrow's load, 15 minutes at a time, from past data, weather and the timetable. If the forecast peak comes close to this year's maximum, facility management gets a warning a day ahead and can stagger the start-up.

What it needs: , and confirmation from facility management that the campus is billed this way.

Requirements

What the system must do, how well it must do it, and the rules we work within. We agree these with the professors and facility management.

What the system does

Priority	Requirement
Must	Collect data from every meter automatically (data acquisition)
Must	Store everything in one place with correct timestamps (central storage, time synchronisation)
Must	Keep collecting during network outages and fill the gap afterwards (store-and-forward buffering)
Must	Dashboard with live and historical values (visualisation)
Must	Data health check (data validation)
Must	Add a new meter through a settings file, without new code (configurability, scalability)
Should	Weekly power-quality report (automated reporting)
Should	Alerts by e-mail or Teams (alerting)
Should	Export data for research (data export, API)

Priority	Requirement
Could	Machine-learning features (ML analytics)

Logins and access

Priority	Requirement
Must	Personal username and password for everyone; no shared accounts (authentication)
Must	Three roles: Admin, Operator (facility management) and Viewer (read only) (role-based access control, RBAC)
Must	All default passwords changed: meters, gateway, database, dashboard (system hardening)
Must	No passwords in code or Git (secrets management)
Must	Each program has its own login with only the rights it needs (least privilege)
Should	Login with existing university accounts (single sign-on, SSO)
Should	Logins recorded; account locked after repeated failed attempts (audit logging, account lockout)

How well it works

- Runs 24/7 without us looking after it.
- Secure: read-only access to the meters, on a separate network.
- Maintainable after we leave: documented, everything in Git, easy to restart.
- Free, open-source tools where possible.

Rules we work within

- **Electrical safety:** only facility management's electricians work on the panels.
- **IT security:** campus IT approves network access.

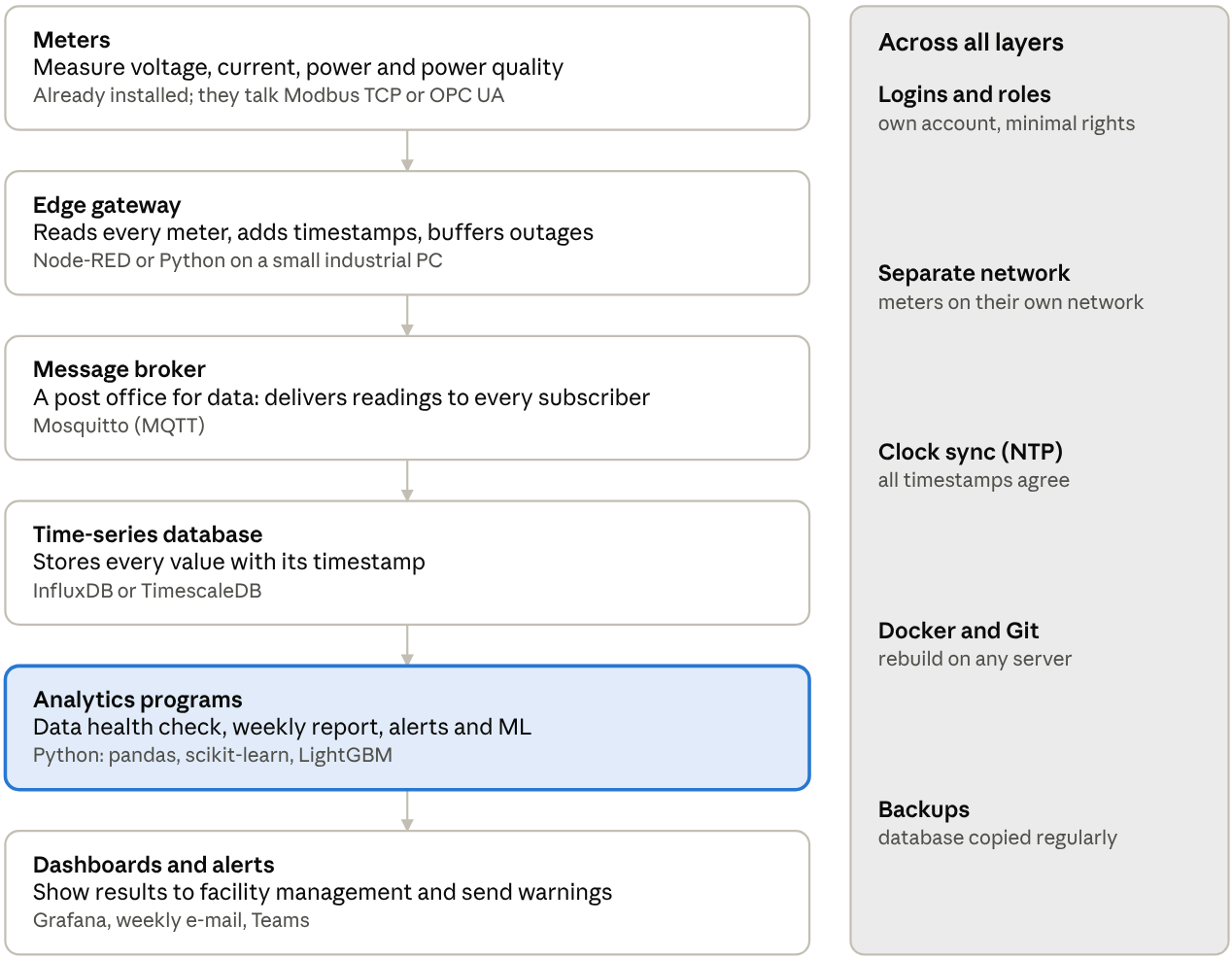
- **Time:** finished by the end of the semester.
- **Budget:** to be clarified.
- **Data protection:** data stays at building or feeder level, not room level.

What we measure

- **Basic:** voltage, current, power, energy, frequency.
- **Power quality:** harmonics, flicker, unbalance, dips and swells. Exactly which depends on the meter model.

System blueprint

Data flows from the meters at the top to the people at the bottom. Each layer does one job, and the analytics layer is where our innovations live.



System blueprint · 6 layers, plus what applies across all of them

Tools for each layer

Layer	Likely tool	Why
Meters	Already installed	They usually talk Modbus TCP or OPC UA
Edge gateway	Node-RED or Python on a small industrial PC or Raspberry Pi	Node-RED is already familiar from the NILM project
Message broker	Mosquitto (MQTT)	Free, and the standard in Industrial IoT

Layer	Likely tool	Why
Database	InfluxDB or TimescaleDB	Built for time-stamped data
Analytics	Python (pandas, scikit-learn, LightGBM)	Covers the reports and all the machine-learning ideas
Dashboards	Grafana	Free, with logins, roles and alerting built in
Running it all	Docker	Each part runs in its own container and restarts with one command

Across all layers

These apply to every layer, not to one box:

- **Logins and roles:** every person and every program has its own account, with only the rights it needs.
- **Separate network:** the meters and the gateway sit on their own network, apart from the campus office network, as agreed with campus IT.
- **Clock sync (NTP):** all devices take their time from one source, so timestamps from different meters line up. The event detective depends on this.
- **Docker and Git:** every part runs in a container, and all code and settings live in Git, so the system can be rebuilt on another server with a few commands.
- **Backups:** the database is copied regularly to a second place, so a broken disk does not lose months of data.

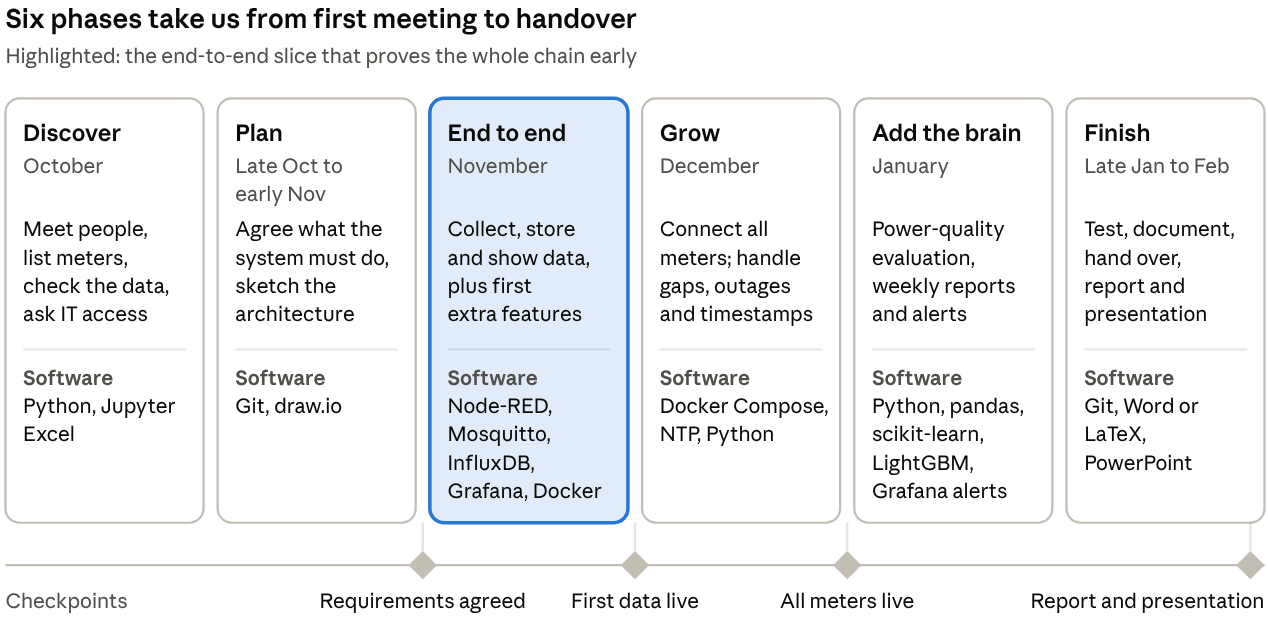
Why build it in layers

- **Each part does one job and can be swapped.** If we change from InfluxDB to TimescaleDB later, the meters and dashboards do not notice.
- **One broken part does not lose data.** If Grafana crashes, data keeps flowing into the database. If the network drops, the gateway buffers readings until it is back.
- **Adding a meter touches only one place:** the gateway's settings file.

Every requirement has a home: store-and-forward lives in the gateway, logins and roles in Grafana, and the data health check in analytics.

Phased roadmap

The six phases from How we will do this project, laid out in time. At each checkpoint we stop and confirm with the professors and facility management before moving on.



Phased roadmap · 6 phases, 4 checkpoints, winter term 2026/27

Checkpoint	When	What done means
Requirements agreed	Early November	The professors and facility management accept the requirements list
First data live	End of November	Data is collected, stored and shown on a dashboard
All meters live	End of December	Every meter is connected, and gaps, outages and timestamps are handled
Report and presentation	February	The system is handed over, and the report and presentation are delivered

Exact dates are still to be agreed with the professors.

Software by phase

Phase	Software	What we use it for
Discover	Python and Jupyter, Excel	Look through the data we were given; keep the list of meters
Plan	Git, draw.io	Set up the shared repository; sketch the architecture
End to end	Node-RED (or Python), Mosquitto (MQTT), InfluxDB (or TimescaleDB), Grafana, Docker	Collect, deliver, store and show the data
Grow	Docker Compose, NTP, Python	Run all parts together, keep clocks in sync, run the data health check
Add the brain	Python (pandas, scikit-learn, LightGBM), Grafana alerting	Weekly reports, alerts and machine learning
Finish	Git, Word or LaTeX, PowerPoint	Documentation, report and presentation

Session roadmap

One learning session per topic, in the same order as the project phases. Each session starts from this doc and ends by adding what we learned to it.

Session	Topic	What we learn	Phase
1 (done)	Vision, ideas and blueprint	Why the project matters, innovation ideas, requirements, blueprint, roadmap	Discover
2	Power-quality basics	What each value means, the EN 50160 limits, how meters build 10-minute values	Discover
3	The campus power system	Transformer, main and sub-distributions, where the meters sit, how to read a meter datasheet	Discover
4	The data we were given	First look in Python and Jupyter: what is in it, first charts, gaps and errors	Discover

Session	Topic	What we learn	Phase
5	Talking to the meters	Modbus TCP or OPC UA; a first live reading with Node-RED or Python	End to end
6	Message broker	MQTT and Mosquitto; how to name topics so the system can grow	End to end
7	Storing the data	Time-series database, data model, how long to keep what	End to end
8	Dashboards and logins	Grafana panels that answer questions; users, roles and passwords	End to end
9	Running it reliably	Docker, data health check, buffering during outages, clock sync, backups	Grow
10	Reports and alerts	The weekly power-quality report and e-mail or Teams alerts	Add the brain
11	Machine learning	Clustering, anomaly detection and load forecasting on our data	Add the brain
12	Finishing	Testing, documentation, handover, report and presentation	Finish

Homework before Session 2

- [ ] Find the meter model or models, and download their datasheets and communication manuals.
- [ ] Open the data we were given and note the time period, how often values are recorded, which values it contains, and the file format.
- [ ] Ask facility management for the campus single-line diagram and where each meter is installed.
- [ ] Book the first meeting with the professors and facility management.
- [ ] Ask campus IT how we can get network access to the meters.

Sources

Papers related to each topic of the project, in the order we meet them. Start with the three marked **read first**.

Power quality and how it is measured

- Bollen, M. H. J. *Understanding Power Quality Problems: Voltage Sags and Interruptions.* IEEE Press, 2000. ISBN 978-0-7803-4713-7. The standard textbook on voltage dips, interruptions and power-quality standards. **Read first.**
- IEC 61000-4-30 Ed. 4.0 (2025). Power quality measurement methods. The standard that defines how each power-quality value is measured, in two classes (A and S). Check which class our meters meet. **Read first.**
- Chawda, G. S. et al. Comprehensive Review on Detection and Classification of Power Quality Disturbances in Utility Grid With Renewable Energy Penetration. IEEE Access 8, 146807–146830 (2020). Why renewables and power electronics cause more disturbances.
- Mahela, O. P., Shaik, A. G., Gupta, N. A critical review of detection and classification of power quality events. Renewable and Sustainable Energy Reviews 41, 495–505 (2015).
- Samanta, I. S., Panda, S., Rout, P. K., Bajaj, M. et al. A Comprehensive Review of Deep-Learning Applications to Power Quality Analysis. Energies 16(11), 4406 (2023).

Collecting data: industrial protocols

- TU Munich. OPC UA versus ROS, DDS, and MQTT. Compares the features, overhead and performance of industrial and IoT protocols, including OPC UA and MQTT.

Storing data: time-series databases

- Liu, R., Yuan, J. Benchmarking Time Series Databases with IoTDB-Benchmark for IoT Scenarios. arXiv, 2019. Compares InfluxDB, OpenTSDB, KairosDB and TimescaleDB.
- Performance Study of Time Series Databases. arXiv, 2022. Compares InfluxDB, TimescaleDB, Druid and Cassandra on loading, storage space and queries.

Data quality: gaps in meter data

- Stojcheska, D. et al. Large-Scale Evaluation of Advanced Imputation Methods for Missing Values in Smart Meter Data. arXiv, 2026. Methods for filling gaps of 1 to 168 hours in meter data.

Showing data: dashboards and visualisation

- Vera-Piazzini, O., Scarpa, M., Peron, F. Building Energy Simulation and Monitoring: A Review of Graphical Data Representation. Energies 16(1), 390 (2023). Which charts suit which energy question and which user.
- Marini, K., Ghatikar, G., Diamond, R. C. Using Dashboards to Improve Energy and Comfort in Federal Buildings. Lawrence Berkeley National Laboratory report. Dashboards as part of a wider energy information system.

Machine learning on energy data

- Himeur, Y. et al. Artificial intelligence based anomaly detection of energy consumption in buildings: A review, current trends and new perspectives. Applied Energy 287, 116601 (2021). Free version on arXiv. Background for smart alerts. **Read first.**
- Miller, C., Nagy, Z., Schlueter, A. A review of unsupervised statistical learning and visual analytics techniques applied to performance analysis of non-residential buildings. Renewable and Sustainable Energy Reviews 81, 1365–1377 (2018). Background for clustering daily load profiles.
- Amasyali, K., El-Gohary, N. M. A review of data-driven building energy consumption prediction studies. Renewable and Sustainable Energy Reviews 81, 1192–1205 (2018). Background for the peak warning.
- Miller, C. et al. The Building Data Genome Project 2, energy meter data from the ASHRAE Great Energy Predictor III competition. Scientific Data 7, 368 (2020). An open dataset for practising before our own data is ready.
- Liu, F. T., Ting, K. M., Zhou, Z.-H. Isolation Forest. IEEE ICDM 2008, 413–422.
- Ke, G. et al. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 30, 3146–3154 (2017).

Law

- EnEfG § 6: energy-saving duties of public bodies.

Some papers are behind a paywall. Try the university library, or a free version where one exists.